

1) CODE OUTPUT:

```
nithya@WIN-482M0QMA856:~$ cc practical1.c
nithya@WIN-482M0QMA856:~$ ./a.out
Enter a Linux command: ls
Parent process
Parent process id is 1799
Child process
My parent process id is 332
Child process id is 1800
Parent process id is 1799
FIFO1          dir5          mybio.txt     practice2.c
OSSP_PROJECT   dir6          myfifo        prodscons.c
OSSP_PROJECTfifo_writer.c  dirz          namedpipe1.c  receiver.c
OSSP_PROJECTpipe_ipc.c    exec1.c      namedpipe2.c  sender.c
OS_PRACTICE    exec2.c      numbers.txt   sender1.c
a.out          execl1.c     open1         shm1.c
a1             f1           open1.c       shm2.c
a10            f2           open2.c       sigint.c
a10.c          f3           open3.c       signal2.c
a7             f4           open4.c       skill1.c
a8             f5           permission_lab skill2.c
a9             f7           pipe           skill3.c
abc            f8           pipe1.c       sleep.c
b1             fifo_writer.c pipe2.c       square.c
b2             linux_task1  pipe3.c       student.txt
backup         linux_task2  pipe4.c       student_portal
bio.txt        linux_task3  pipe5.c       xyz
data.txt       linux_task4  pipe_ipc.c    z7
dir1           malloc       practical1.c  zombie1.c
dir2           malloc.c     practical2.c  zombie2.c
dir3           malloc_asan  practical4.c
dir4           my_fifo     practice1.c
Child process completed
nithya@WIN-482M0QMA856:~$ |
```

B) LINUX COMMANDS:

```

hithya@WIN-482M0QMA856:~$ uname
Linux
hithya@WIN-482M0QMA856:~$ lscpu
Architecture:            x86_64
CPU op-mode(s):          32-bit, 64-bit
Address sizes:            48 bits physical, 48 bits virtual
Byte Order:               Little Endian
CPU(s):                   16
On-line CPU(s) list:      0-15
Vendor ID:                AuthenticAMD
Model name:               AMD Ryzen 7 7730U with Radeon Graphics
CPU family:               25
Model:                    80
Thread(s) per core:       2
Core(s) per socket:       8
Socket(s):                 1
Stepping:                  0
BogoMIPS:                  3992.39
Flags:                    fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge m
ca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall
nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep
_good nopl xtopology tsc_reliable nonstop_tsc cpuid ex
td_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16
sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand h
ypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a
misalignsse 3dnowprefetch osvw topoext perfctr_core ss
bd ibrs ibpb stibp vmmcall fsgsbase bmi1 avx2 smep bmi
2 erms invpcid rdseed adx smap clflushopt clwb sha_ni
xsaveopt xsavec xgetbv1 xsaves clzero xsaveerptr arat
npt nrip_save tsc_scale vmcb_clean flushbyasid decodea
ssists pausefilter pfthreshold v_vmsave_vmload umip va
es vpclmulqdq rdpid fsrm
Virtualization features:
Virtualization:           AMD-V
Hypervisor vendor:        Microsoft
Virtualization type:       full
Caches (sum of all):
L1d:                       256 KiB (8 instances)
L1i:                       256 KiB (8 instances)
L2:                         4 MiB (8 instances)
L3:                        16 MiB (1 instance)
NUMA:
NUMA node(s):              1
NUMA node0 CPU(s):         0-15

```

```

nithya@WIN-482M0QMA856:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G   0 disk [SWAP]
sdd   8:48   0    1T   0 disk /mnt/wslg/distro
/

nithya@WIN-482M0QMA856:~$ ps
  PID TTY          TIME CMD
  364 pts/0    00:00:00 bash
 1980 pts/0    00:00:00 ps

nithya@WIN-482M0QMA856:~$ top
top - 13:33:22 up 59 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  25 total,   1 running, 24 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  6849.6 total,  6095.2 free,   609.1 used,   299.6 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used,  6240.5 avail Mem

   PID USER      PR  NI   VIRT   RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24284   15372  11516 S   0.0   0.2   0:01.89 systemd
    2 root        20   0    3180    2212   2072 S   0.0   0.0   0:00.02 init-syste+
    6 root        20   0    3180    2040   1940 S   0.0   0.0   0:00.00 init
   52 root        19  -1   50400   16764  15464 S   0.0   0.2   0:00.54 systemd-jo+
   69 systemd+    20   0   22416   14444  11944 S   0.0   0.2   0:00.18 systemd-re+
   91 root        20   0   35260   12288   9176 S   0.0   0.2   0:03.17 systemd-ud+

```

Command

Role/Purpose

uname	Displays the os being used.
lscpu	Displays CPU architecture, cores, threads, and processor information.
lsblk	Displays storage devices, partitions, and their sizes.
ps	Displays currently running processes and their PIDs.
top	Provides real-time information about processes and CPU/memory usage.